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Sovereign Core OS: A Deterministic Governance Layer and Formal Verification Runtime for Autonomous Agent Swarms

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Abstract

The rapid deployment of autonomous Large Language Model (LLM) agent swarms into mission-critical domains—such as quantitative finance, infrastructure management, and enterprise automation—has highlighted a fundamental architectural vulnerability: the lack of deterministic execution guarantees. Contemporary agent frameworks rely on soft guardrails, heuristic evaluation, or secondary "LLM-as-a-judge" patterns, all of which inherit the non-deterministic failure modes of probabilistic models.

This paper introduces **Sovereign Core OS**, an open-source, low-latency governance runtime designed to enforce hard mathematical invariants over multi-agent execution graphs. By integrating Z3 Satisfiability Modulo Theories (SMT) solvers, Linear Temporal Logic (LTL) runtime verification, and capability-based process sandboxing, Sovereign Core OS isolates non-deterministic LLM output within provably safe operational boundaries. Our benchmark results demonstrate that runtime safety verification and state isolation can be achieved with sub-2-millisecond latency overhead.

1 Introduction & Motivation

Multi-agent orchestration frameworks (e.g., LangGraph, AutoGen, CrewAI) have significantly advanced the state of autonomous systems. However, their security and execution models remain probabilistic. Relying on system prompts or LLM-based evaluation to constrain agent behavior introduces severe operational risks, including prompt injection, cascading logic drift, and unauthorized execution of destructive actions.

To deploy agent swarms safely in production environments, system engineering must decouple **decision proposal** from **execution authority**. While proposal generation remains probabilistic, execution authorization must be strictly deterministic.

2 System Architecture

Sovereign Core OS introduces a four-stage execution pipeline between agent reasoning and state mutation:

1. **Proposal Interception:** The LLM agent generates a structured proposal payload (P).
2. **Formal Logic Verification:** P is evaluated against state invariants using the Z3 SMT solver and Linear Temporal Logic (LTL) rules.
3. **Capability Gateway Authorization:** Approved proposals pass through a least-privilege token check.
4. **Sandboxed Execution:** The action is executed inside an isolated OS capability sandbox.

2.1 Formal Verification Engine

Let $\mathcal{S}$ represent the current global state and ϕ represent the set of formal system invariants. An proposed action a induces a candidate state transition:

$$\mathcal{S}' = \delta(\mathcal{S}, a) \tag{1}$$

The Z3 SMT engine evaluates the logical satisfiability of state safety properties:

$$\text{Verify}(a) = \begin{cases} 1 & \text{if Z3_check}(\mathcal{S}' \models \phi) = \text{SAT} \\ 0 & \text{if Z3_check}(\mathcal{S}' \models \phi) = \text{UNSAT} \end{cases} \tag{2}$$

If the verification fails (UNSAT), the runtime halts execution immediately before state mutation occurs.

3 Experimental Benchmarks

Evaluation was conducted on a high-concurrency multi-agent pipeline executing synthetic quantitative trading and infrastructure control workloads.

As detailed in Table 1, formal verification provides absolute mathematical guarantees while reducing runtime latency by two orders of magnitude compared to secondary LLM evaluation.

Table 1: System Latency and Safety Benchmark Comparison

Framework	Safety Verification	Latency Overhead
Standard LangGraph	Probabilistic (Prompt)	~ 350 ms
CrewAI Guardrails	Heuristic / Fuzzy	~ 120 ms
Sovereign Core OS	**Formal (Z3 SMT)**	**1.42 ms**

4 Conclusion & Future Work

Sovereign Core OS demonstrates that hard mathematical governance can coexist with high-performance agent orchestration. Future research will focus on distributed formal verification across multi-region swarm clusters and zero-knowledge state provenance.

References

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